**Code No: 136CD** **R16**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech III Year II Semester Examinations, August/September - 2024**
**INFORMATION SECURITY MANAGEMENT**
**(Computer Science and Engineering)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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**PART – A**

**(25 Marks)**

1.a) What is a Common Vulnerability and Exposure (CVE)? [2]
b) Discuss the characteristics of Man-In-The-Middle (MITM) attacks. [3]
c) What are the different states of information? [2]
d) What is the role of access control mechanisms in securing networks and information? [3]
e) What are the common sources and causes of data leakage within organizations? [2]
f) What are Key Performance Indicators (KPIs) related to data leakage prevention? [3]
g) Define security policy implementation. [2]
h) What metrics can be used to measure the effectiveness of information security policy implementation? [3]
i) Mention the security roles in information security management. [2]
j) What is the risk analysis process in information security management? [3]

**PART – B**

**(50 Marks)**

2. What is information security, and why is it important in today's digital age? [10]

**OR**

3. Describe the categories of cyber-attacks based on their objectives (e.g., denial of service, data breach). [10]

4. Discuss the role of DNS (Domain Name System) in translating domain names to IP addresses. [10]

**OR**

5. Explain the concepts of confidentiality, integrity, and availability (CIA) in relation to critical information. [10]

6. Describe the financial and reputational consequences of data leakage for organizations. [10]

**OR**

7. Explain the importance of employee training and awareness programs in mitigating data leakage risks. [10]

8. Explain the role of senior management in endorsing and enforcing information security policies. [10]

**OR**

9. Describe the role of security standards and guidelines in shaping organizational security posture. [10]

10. Explain the role of security administrators and analysts in maintaining information security. [10]

**OR**

11. Explain the post-incident analysis and lessons learned process in improving incident response capabilities. [10]

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